Response to Referee Reports

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Preferences for Warning Signal Quality: Experimental Evidence

Games and Economic Behavior · August 31, 2026

We thank the associate editor and both referees for their thorough and constructive reviews. The revised manuscript is stronger for it. Below we address each point in turn. Referee comments are shown in shaded boxes (blue for Referee 1, teal for Referee 2); our responses follow in plain text.

Referee 1

We thank Referee 1 for the careful reading of both the original submission and the revision.

R1.1 Major Comment: Incorrect and Inconsistent Definitions of FP and FN Costs

> The paper now emphasizes the distinction between false-positive/false-negative rates and false-positive/false-negative costs. This is a useful clarification. However, the definitions given in the model and in Section 5.2 are not correct. The paper defines FP and FN costs on page 6 as $\pi(1 - P_{01})c$ and $\pi P_{01}L$, respectively, and states that WTP is equally affected by both. There are two problems.
>
> First, the FP cost definition is incorrect. The expression $\pi(1 - P_{01})c$ is the expected cost of true positive protection (correctly protecting in the adverse state), not the cost of false positives. The FP cost should be the expected cost of unnecessary protection when there is no adverse event: $(1-\pi)P_{10}c$. This is consistent with equation (4), where the derivative of risk-neutral WTP with respect to the FP rate is $-(1 - \pi)c$.
>
> Second, there is a subtlety in the FN cost definition. The paper defines it as $\pi P_{01}L$, which is the gross expected loss from missed detections. An alternative definition is $\pi P_{01}(L - c)$, which captures the incremental cost relative to the correct action: under a true positive signal the agent would have protected and paid c, so the net cost of missing the adverse state is $L - c$, not L. This alternative is more consistent with equation (5), where the derivative of WTP with respect to the FN rate is $-\pi(L - c)$.
>
> This matters for Table 5 and the main regression. The dependent variable is WTP minus the risk-neutral value, and the independent variables are labeled FP and FN costs. The paper should explicitly define the variables used in the regressions, show how they map into the theoretical WTP formula (equation 3), and verify that the definitions do not affect the reported results.

Response: [Your response here.]

R1.2 Major Comment: The Interpretation of Table 5 and Figure 5 Should Be Stated More Precisely

I found that the interpretation and conclusion of the paper are not clear and somewhat misleading. For instance, the abstract states that subjects "disproportionately reduce their demand for signals with high false negative rates for rare events." This characterization is imprecise and risks confusion with what the data actually show in subsection 5.2 (conditional on the results are correct).

Under the null hypothesis that subjects are risk-neutral expected-utility maximizers, the benchmark WTP already incorporates FP and FN costs, so all coefficients in the discrepancy regression (Table 5) should be zero. Positive coefficients indicate deviations from the benchmark: as error costs rise, subjects' WTP does not fall enough. The actual finding of the regression table is that subjects selectively under-react to the cost type that the theory identifies as dominant: for low priors, the coefficient on FP costs is large and significant while the coefficient on FN costs is near zero; for high priors, the pattern reverses, with FN costs generating the significant positive coefficient. The theory (equations 4–5) predicts that FP costs should be the primary concern for rare events, since $(1 - \pi)$ is large and most positive signals will be false alarms, while FN costs should dominate for frequent events, since π is large and missed detections become costly. This implies that subjects under-react to precisely the cost type that theory says should matter most:

- Low priors $\Rightarrow$ FP costs should dominate $\Rightarrow$ subjects fail to adjust for FP costs
- High priors $\Rightarrow$ FN costs should dominate $\Rightarrow$ subjects fail to adjust for FN costs

Figure 5 corroborates this from a different angle: when raw WTP is regressed on FP and FN rates by prior, the empirical sensitivity coefficients are roughly equal and flat across priors, whereas theory predicts sharp divergence. The finding is not that subjects overpenalize FN rates for rare events in absolute terms, but rather that they fail to differentially weight FP and FN rates as the prior shifts, producing relative over-sensitivity to FN (undersensitivity to FP) for low priors and the reverse for high priors. I think this is stronger than saying simply that subjects are "asymmetrically under-responsive by prior," but weaker than saying that subjects generally fail to distinguish FP and FN errors. The evidence shows that subjects do not process FP and FN rates in the benchmark Bayesian/EU way. It does not uniquely identify the cognitive mechanism.

The abstract and conclusion should be rewritten to state this more precisely. The paper should also be consistent about whether claims refer to error rates, expected costs, or deviations from the risk-neutral benchmark. The paper would also benefit from using "false alarm" and "missed detection" alongside the FP/FN notation, especially when discussing costs. These terms make it immediately clear why low-prior environments make false alarms more important and high-prior environments make missed detections more important.

Response: XXX Rewrite such that we do not claim subjects don't know FP/FN errors. "'Subjects to account the effect of priors on sensitivity of expected costs to FP/FN rates"'

R1.3 Major Comment: The Mechanism Claim Should Be Softened

> The discussion in Section 5.3 is much improved. The authors now consider risk preferences, probability weighting, anchoring, non-instrumental information, and anticipation of biased protection decisions. The additional BE and IP analyses are also useful. However, I remain unconvinced that the data uniquely identify a failure to distinguish false-positive and falsenegative errors. The evidence is interesting: some subjects report using the proportion of dishonest gremlins as a decision rule, and the BE/IP analyses show anomalous responses to FP and FN rates even when those rates should not affect the posterior given the observed signal (Tables 6–7).
>
> However, the mechanism is not cleanly identified. The same pattern could reflect a broader total-error-rate heuristic, a "dishonest gremlin" heuristic induced by the experimental framing, probability weighting, noisy Bayesian reasoning, task complexity, or some combination. The authors themselves acknowledge that probability weighting may play a role for low priors. More importantly, the leap from "subjects react to irrelevant error rates in BE and IP tasks" to "subjects conflate FP and FN when valuing signals" requires more thorough discussion and the implicit assumption that the same cognitive failure operates across tasks.
>
> A more defensible conclusion would be: under this experimental representation, many subjects appear to use a simplified total-error or total-dishonesty heuristic rather than separately processing how FP and FN rates interact with the prior probability. This interpretation fits the subject explanations and the BE/IP evidence better than the stronger claim that subjects generally cannot distinguish FP and FN errors. It is still an interesting finding.
>
> Relatedly, Section 4 should not be framed only as reassurance that subjects understood the task. The BP, IP, and BE patterns are part of the behavioral phenomenon. The data suggest that subjects respond directionally to incentives and information, but also display substantial heterogeneity, mistakes, and simplified reasoning. That framing would make the transition to the WTP results more natural.

Response: XXX Rewrite to weaken the claims, follow suggestion XXX

R1.4 Minor Comment: Proposition 2

> Ensure the wording explicitly states that the proposition concerns the ratio of sensitivities $(\partial b/\partial P_{01})/(\partial b/\partial P_{10})$, not absolute levels. A brief economic interpretation would help: compared to risk-neutral agents, risk-averse agents are relatively more intolerant of missed detections (false negatives) but more forgiving of false alarms (false positives).

Response: [Your response here.]

R1.5 Minor Comment: Appendix B Proof Numbering

The main text contains two propositions, but the appendix has three proofs (B.1–B.3). The proof for Proposition 2 (ratio of sensitivities) appears in B.3. Section B.2 appears to prove a result for imprudent agents that no longer corresponds to a proposition in the main text. B.2 should be dropped or the main text should be updated to match.

Response: We have now dropped the old Section B.2 that had a proof that is no longer relevant. Thank you for pointing this out.

R1.6 Minor Comment: Appendix B.3 Algebra Error

In equations (13) and (14), the denominator of the last expression is $(1-\pi)L$. It should be $(1-\pi)c$. This can be verified by simplifying the middle expression:

$$\frac{\pi}{1-\pi} \cdot \frac{u'(\xi_1)(L-c)}{u'(\xi_2)c} = \frac{\pi(L-c)}{(1-\pi)c} \cdot \frac{u'(\xi_1)}{u'(\xi_2)}.$$

Response: We thank the referee for catching this typo. We have corrected the denominator in equations (13) and (14) from $(1-\pi)L$ to $(1-\pi)c$. The stated result is unaffected.

R1.7 Minor Comment: Cross-Reference Error

Page 26 references "Appendix Table A10," but the appendix table with matching content (WTP deviations with IP bias controls) is numbered A8. The table numbering appears to be off.

Response: [Your response here.]

R1.8 Minor Comment: Typos

- Page 7: "Equations 6-s 7" → "Equations 6–7."
- Page 15, footnote 10: missing comma between the two parameter estimates ($\hat{\pi} = 0.43$ $\hat{\theta} = 0.25$).
- Page 18: "the value for for a prior of 0.2" — duplicated word.
- Page 27: "the observed sensitivity patterns emerges" → "emerge."
- Page 27: "characteristcs" → "characteristics."

Response: All five typos have been corrected. We thank the referee for the careful reading.

Referee 2

We thank Referee 2 for the detailed and constructive comments on the revision.

R2.1 Treatments and Treatment Assignment

(a) While the new "Treatments and Treatment Assignment" subsection on page 10 provides additional detail on how treatments are assigned, it remains incomplete. It states that each subject is randomly allocated to a pair of priors but does not explain how information structures are assigned.

(b) The text also notes that "the first treatment was always the one with a fully truthful information structure," which suggests that information structures are not randomly assigned across subjects. For transparency and to avoid confusion, the authors should provide at least a general description and justification of how information structures are allocated.

(c) Because information structures do not appear to be randomly assigned, the resulting sample sizes across treatment cells (i.e., prior × information structure combinations) are likely to differ. I encourage the authors to report sample sizes by treatment cell. Given the large number of cells ($4 \times 12 = 48$), a table in the appendix would likely suffice.

Response: Thank you for the suggestion. We have now added the full lists of treatments and sequences in the appendix. We have also revised the description of the treatment assignment in Section 3 as follows:

> **Treatments and Treatment Assignment.** Table XX summarizes our treatments for tasks other than BP. The treatments combine four priors with 12 different information structures for a total of 48 treatments. These treatments allow for a distribution of posteriors (for black and white signals) that spans a wide range of interior values and allows for a wide range of theoretical WTP values (Appendix Figure XX).
>
> Based on these 48 treatments, we constructed 32 unique treatment sequences that were randomly assigned to subjects. Each sequence comprises six rounds of treatments that combined a pair of priors of either (0.1, 0.3) or (0.2, 0.5) with three information structures. Within a sequence, treatments were presented by prior; in half of the sequences, the first prior is larger than the second. For each prior, the first treatment was always the one with a fully truthful information structure. The order of priors and signals stays constant across tasks, but vary between sequences. Appendix Tables A1 and A2 present the full lists of treatments and sequences.

R2.2 Description and Groupings in Tables 2, 3, and A4

(a) On page 12, the text states that "Table 2 presents the average protection decisions by prior and signal type," but Table 2 does not, in fact, report descriptive statistics by

prior. Instead, it groups observations by signal value (white/black) and by information-structure type (truthful, FP-only, FN-only, both errors). The same mismatch appears in the caption of Appendix Table A4, which refers to "signal type and prior," although the table is actually organized by signal value and informationstructure type.

(b) Relatedly, the authors could briefly justify why Table 2 groups observations by signal value and information structure while pooling across priors. A single sentence would suffice. Providing this explanation would improve transparency and clarity for readers, especially since the same grouping is used in Tables 3 and A4.

Response: Typo, we moved the tables to the appendix. Adjust the text in the paper.

R2.3 Claim on Page 16 Regarding the Introduction of the FN Rate (Row 1 vs. 2)

(a) The paragraph below Table 3 states that introducing the FP rate (row 1 vs. 3 of Table 3) magnifies the upward updating bias for white signals—which is correct: the Bayesian posterior remains at 0, while the mean updating error increases from 0.048 to 0.210. The text then claims that "we find a similar effect for the introduction of the FN rate (row 1 vs. 2)." However, comparing row 1 and row 2, the mean updating error does not appear to change significantly (0.048 vs. 0.051).

(b) Moreover, introducing the FN rate changes the Bayesian posterior for the white signal (from 0.000 to 0.119), so the comparison between row 1 and row 2 does not hold the posterior fixed—unlike the comparison between row 1 and row 3.

Response: Correct the text

R2.4 Section 4's Closing Summary Understates the Evidence in Tables 2 and 3

(a) The paragraph just before Section 5 (page 17) concludes that "All of this suggests that subjects understand the experimental environment well enough to allow us to draw reasonable inferences from our WTP elicitation." While I agree that subjects appear to understand the environment sufficiently well, this framing understates several important findings from Section 4.

(b) For example, Tables 2 and 3 show that increasing the FP (FN) rate affects protection decisions and updating errors even in cases where the rate should not influence the Bayesian posterior (e.g., FP for a white signal in Table 3, row 1 vs. 3). These patterns provide suggestive evidence of the FP/FN-confusion mechanism. I would encourage the authors to add a sentence foreshadowing that the descriptive analysis already reveals patterns that are later revisited and explained in Section 5.3. This would make the main mechanism visible from Section 4 onward, rather than only at the end of the paper.

Response: Add it back in foreshadowing this result.

R2.5 Hypothesis 1 and Result 1 Overstate What the t-Tests in Table 4 Establish

(a) Hypothesis 1 (page 17) states that "subjects' WTPs for signals are equal to their value for risk-neutral agents," and Result 1 concludes that "on average, there are no significant discrepancies between the empirical WTP and the risk-neutral benchmark." However, the supporting analysis relies on a t-test of the mean of $\Delta b_{is} = b_{is} - b_s^*$. Failing to reject $E[\Delta b_{is}] = 0$ is a much weaker result than establishing $\Delta b_{is} = 0$, as large but offsetting over- and under-valuations across subjects and treatments may cancel out in the mean. Given the likely presence of individual heterogeneity, this distinction is important. I suggest one of the following revisions:

i. Restate Hypothesis 1 and Result 1 more precisely, for example: "subjects' WTPs are not systematically above or below the risk-neutral benchmark on average," which is itself an informative finding; or

ii. Test the stronger claim directly, for instance by examining whether the absolute deviation $|\Delta b_{is}|$ is significantly different from zero.

Response: Do (i), explain why (ii) is not necessarily doable

R2.6 (Optional) Reorganize Section 5.3 to Give Greater Prominence to the FP/FN-Confusion Mechanism

(a) Given that the FP/FN-confusion mechanism is the primary mechanism, it might be preferable to present it as the first candidate explanation rather than the last, and then proceed to evaluate and rule out alternative mechanisms.

Response: Reorganize to follow suggestion

R2.7 Risk-Preference Discussion (pp. 23–24)

In the risk-preference discussion (pp. 23–24), the authors could also note that the main effects of FP and FN costs remain significant, rather than focusing solely on the insignificance of the interaction terms.

Response: Follow

R2.8 Probability Weighting Discussion (pp. 24–25)

(a) In equation 9 and the definition of Λ, the numerator should be P_{01} instead of P_{00}?

(b) I appreciate the authors' careful argument with rank-dependent utility! In my previous report, I was meant to refer to the probability weighting function from prospect theory (Kahneman and Tversky, 1979). But the arguments turn out to be quite similar because the particular payoff set used in the experiment.

(c) The flow of the argument is clear, and the observation that "$\pi + (1 - \pi)P_{10} \leq 0.75$, meaning that the value is still likely to fall in the mid-range of an S-shaped weighting

function," is insightful. However, this depends on the rotation point of the probability weighting function. Theoretically, if the rotation point lies far to the left, then $\pi + (1 - \pi)P_{10} \leq 0.75$ might still fall within the steeper region of the weighting function (such that the denominator is large and Λ is small, implying lower relative sensitivity). It would be helpful if the authors could provide evidence for "middle" rotation points in existing studies (e.g., Gonzalez and Wu (1999), as cited). That said, since the shape of the weighting function is not restricted, this possibility may be too flexible to rule out. In any case, it would be helpful to discuss this issue more thoroughly.

Response: [Your response here.]

R2.9 The "Anticipation of Bias in Protection Decision"

(a) Table A10 should be Table A8; the caption of Table A8 should refer to WTP deviations rather than simply WTP.

(b) More importantly, I am not entirely convinced that the two newly introduced variables adequately test the mechanism of "anticipation of bias in protection decisions." If I understand correctly, Δprot measures the extra cost incurred by the subject relative to the optimal benchmark, while Δloss measures the additional loss incurred relative to the same benchmark. The coefficients in Table A8 are negative, suggesting that greater mistakes (in terms of cost or loss) reduce WTP deviations. In other words, subjects who perform worse in the IP task appear to be more accurate in their WTP, which may indicate that they account for their own mistakes.

(c) Given that the coefficients on the two new variables are statistically significant, the conclusion that "subjects' biases in how they will utilize signals when facing an IP task do not explain the observed pattern of sensitivities to FP and FN rates" seems premature. A more appropriate interpretation is that the coefficients on FP and FN do not appear to change after introducing these two variables.

(d) However, a more informative regression specification would include interaction terms between the mistake variables and FP/FN rates. The intuition is that, if anticipation of bias is the primary channel, then mistakes in protection decisions should mediate the relationship between WTP deviations and FP/FN. That is, if subjects expect to make more mistakes in the IP task at higher FP/FN rates, they may reduce their WTP, leading to larger deviations. Along similar lines, it may be preferable to distinguish mistakes in BP and IP directly, rather than separating them into cost and loss components.

(e) Finally, even if anticipation of bias is an important (or primary) channel, it does not undermine the FP/FN-confusion mechanism; if anything, it reinforces it. Tables 2 and 7 already show that subjects' IP protection decisions respond to FP and FN rates in ways consistent with FP/FN confusion. For instance, subjects choose to protect upon receiving a white signal even when there are no false negatives but only false positives (Table 7 provides related evidence). Anticipation of bias suggests that this confusion arises not only at the WTP stage but also during the IP stage, with such anticipation

leading to larger deviations of WTP from the risk-neutral optimal benchmark.

Response: [Your response here.]

R2.10 Cross-Task Consistency Analysis

The authors should consider incorporating the cross-task consistency analysis from the response letter (Table 2 on page 13) into the paper, or at least including it in the appendix.

Response: Move it into the appendix

R2.11 (Optional) The "Dishonest Gremlin" Framing

I understand the authors' argument that re-running the experiment with a different framing of the information structure might make pooling unjustifiable. However, the authors could consider adding a brief discussion of this issue (perhaps at the end of the paper). Even without the "dishonest" framing, individuals may still conflate the two types of errors (FP and FN). It would therefore be helpful for the authors to acknowledge and discuss this potential limitation.

Response: Add a discussion + a footnote why we didn't incorporate the neutral frame because of pooling concerns.

R2.12 Exposition

(a) On page 11, the sentence "this section also provides ... the theoretical predictions" is a tad imprecise: the predictions refer to those of the risk-neutral benchmark, not theoretical predictions in general. I would suggest "... from the theoretical predictions of the risk-neutral benchmark" (or similar) for clarity.

(b) Relabel "Posterior" as "Bayesian posterior" in Tables 2, 3, A2, and A4. It would help the readers in case they mistake it as elicited posterior.

(c) In both Table 5 and Table A8, the boundary between the main regression panel and the lower panel reporting the risk-averse and risk-loving subgroup tests is not easy to spot at a glance. A horizontal rule separating the two would make the structure of these tables much easier to spot.

(d) (Optional.) The appendix tables would be easier to navigate if the captions (or notes) pointed the reader to the coefficient or column that matters for the corresponding discussion in the main text. A short closing sentence in each caption — e.g., "comparing the coefficient of XYZ and ABC in column 2 suggests that ..." — would substantially reduce the cost of cross-referencing.

(e) On page 21, the text refers to "the systematic WTP biases reported above" without specifying which biases are meant. I would suggest the authors to be more precise here.

Response: XXX FOLLOW AFTER MAIN ANALYSIS DONE XXX

R2.13 Typos

> - There is an extra "=" sign in equation (2) on page 5.
> - On page 7, "Equations 6-s 7" should read "Equations 6 and 7" (or similar).
> - Footnotes 10 and 11 are missing a full stop at the end.
> - Footnote 13 (page 21): "average these errors across all the decisions with identical priors, false positive, and false negative rates". Should "identical" be "different"?
> - In the last paragraph on page 26, Table A10 should be Table A8.

Response: All five typos have been corrected. We thank the referee for the careful reading. XXX MAKE SURE WE DO THIS XXX